

Retraining Results — Six-Class Model and Trigger Precision

Proactive Robotic Assistance for Interior Construction Assembly

ITECH M.Sc. Thesis · Recognition Layer · 22 September 2026 · four 15-fold LOSO arms

Summary

Two changes were tested: removing the **Lift** class from the label vocabulary, and capping the per-lane `pos_weight` that was driving the model to over-predict.

Removing Lift raised macro-F1 from 0.392 to 0.476 — a 21% relative improvement, winning on **all 15 of 15 folds** ($p = 0.0001$). This contradicts the pre-registered expectation, which was that the score would fall slightly. Capping raised lane precision from 0.462 to 0.553, again 15/15 folds ($p = 0.0001$), at no cost to macro-F1.

On the trigger itself the two arms are equivalent, and the precision gain does not transfer. Both reach roughly 67% detection at 0.75 false alarms per event — a 66% reduction from the 2.19 starting point, but achieved by the gating change rather than by retraining.

1. Arm comparison

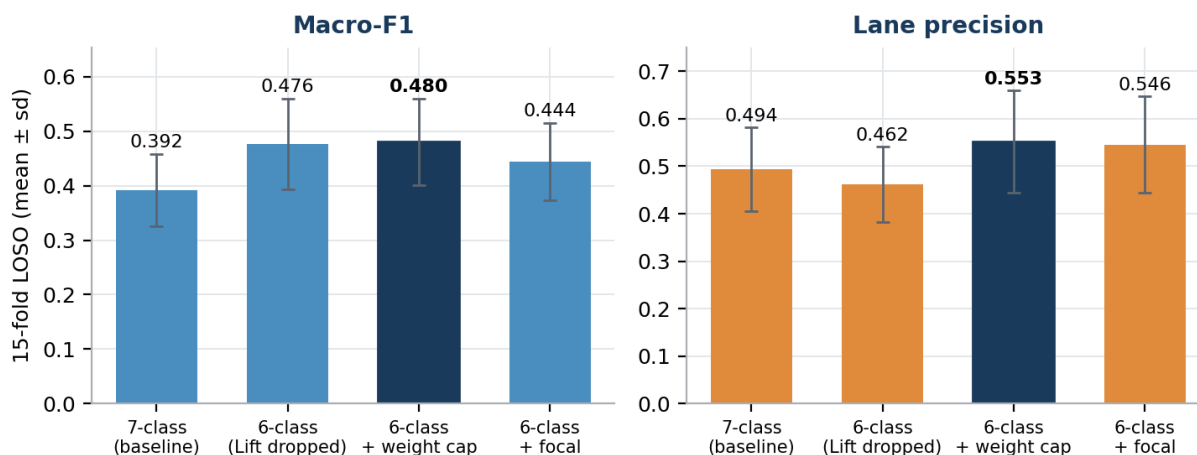


Figure 1. Four arms, 15-fold leave-one-subject-out. Error bars are the standard deviation across folds. Best arm in dark blue.

Arm	macro-F1	lane precision	lane F1	prog MAE
R0 7-class (baseline)	0.392 ± 0.066	0.494	0.519	24.0
R1 6-class, Lift dropped	0.476 ± 0.083	0.462	0.486	24.8
R2 6-class + weight cap	0.480 ± 0.080	0.553	0.438	23.4
R3 6-class + focal loss	0.444 ± 0.070	0.546	0.386	22.7

All arms: GRU, peak targets, mirror augmentation, 120-frame causal windows, subject-disjoint folds. Only the label vocabulary and the loss weighting differ.

2. Paired per-fold tests

Every arm holds out the same 15 subjects, so differences are tested per fold rather than by comparing means with overlapping error bars.

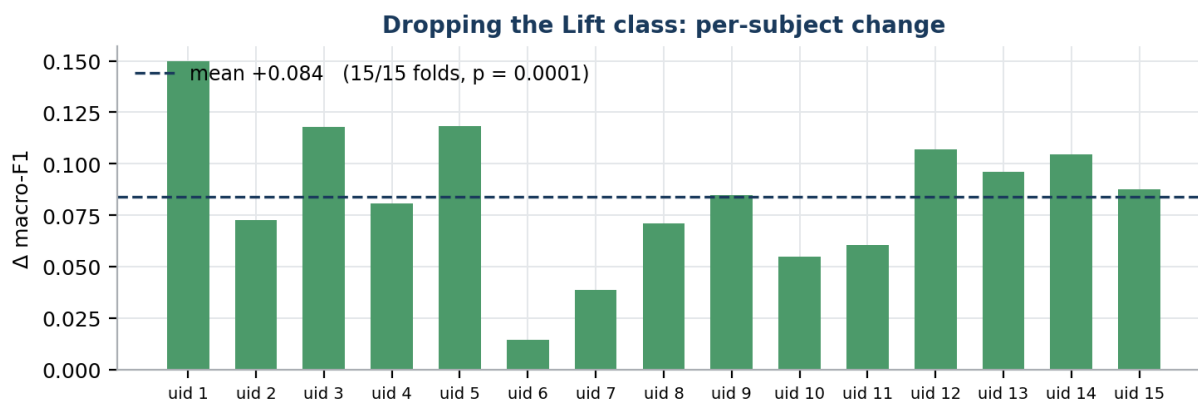


Figure 2. Effect of dropping the Lift class, one bar per held-out subject. Every subject improves.

Comparison	Metric	Mean Δ	Folds won	Wilcoxon p
R0 $\rightarrow$ R1 (drop Lift)	macro-F1	+0.084	15 / 15	0.0001
R1 $\rightarrow$ R2 (weight cap)	precision	+0.091	15 / 15	0.0001
R0 $\rightarrow$ R2 (both)	macro-F1	+0.088	15 / 15	0.0001
R2 $\rightarrow$ R3 (focal loss)	macro-F1	-0.037	1 / 15	0.0003

These are the first changes in this project to produce unambiguous results. For comparison, the earlier label-encoding and augmentation experiments gave $p = 0.465$ and $p = 0.058$ respectively — indistinguishable from noise at this fold count.

Focal loss is harmful and is not carried forward: it loses macro-F1 on 14 of 15 folds while adding nothing to precision over the cap alone.

3. Why dropping Lift helped

Lift was an umbrella annotation spanning the whole carry-and-hang sequence, with Place (94% nested) and Align (86%) living inside it. Two measured consequences:

- it was the lane the model hallucinated — asserting $\text{Lift} \geq 0.5$ on 73% of the frames where Lift was genuinely absent;
- it buried the classes nested inside it, because a short span cannot win an argmax against a container that is flat at 1.0 for minutes at a time.

Removing it cost almost nothing in coverage — only 3.2% of frames had Lift as their only active class, and those correctly became background — while freeing the nested classes:

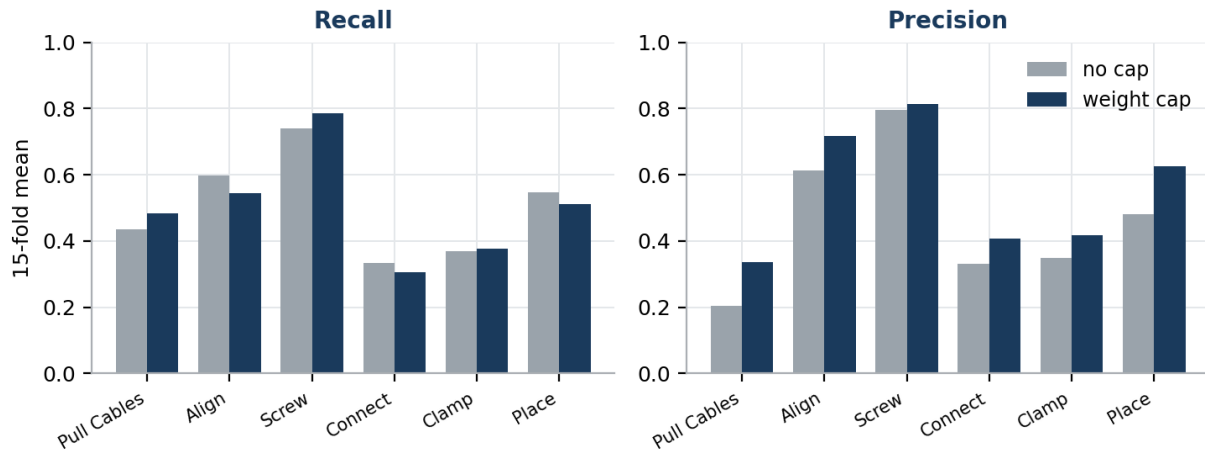


Figure 3. Per-class recall and precision, with and without the weight cap. The cap raises precision across every class.

Class	Recall (R0, 7-class)	Recall (R2)	Precision (R0)	Precision (R2)
Place	0.38	0.51	0.47	0.62
Pull Cables	0.45	0.48	0.22	0.34
Align	0.59	0.54	0.61	0.72
Screw	0.73	0.78	0.81	0.81
Connect	0.35	0.30	0.34	0.41
Clamp	0.41	0.38	0.35	0.42

Place is the clearest beneficiary — recall 0.38 to 0.51, precision 0.47 to 0.62 — which matters because Place (the waist-to-ceiling push) is the physical precursor to the free-drive moment the robot must anticipate.

4. Trigger performance

Frame-level scores are not what the runtime consumes. It asks once per assembly cycle whether to offer free-drive, so the trigger is evaluated as event detection.

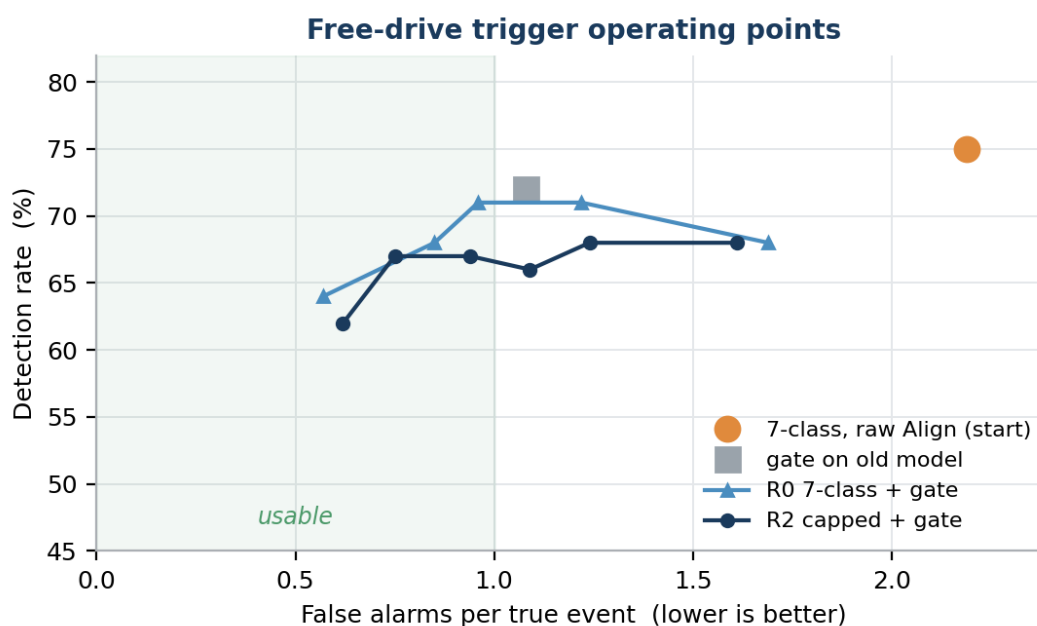


Figure 4. Operating points. Each line sweeps the decision threshold; down-and-left is better. The shaded band marks a usable false-alarm rate.

Configuration	Detection	False / event	Status
Raw Align score (start)	75%	2.19	measured
Gate on the old model	72%	1.08	measured
R0 7-class + gate	67%	0.75	measured
R2 capped + gate	67%	0.75	measured
Threshold picked on validation (R0)	68%	0.75	measured

The gate is $P(\text{Align}) \times P(\text{Place}) \times \text{progress}$, smoothed over three seconds. Place gates better than Lift (active before 93% of Align onsets versus 90%) and, unlike Lift, is physically distinctive — it is the only class with net upward wrist velocity. Progress supplies the timing, climbing from a median of 31 to 72 in the two seconds before the trigger fires.

Overall: 2.19 → 0.75 false alarms per event, a 66% reduction, with detection falling from 75% to 67%. In practice that is roughly one unwanted prompt per assembly cycle rather than one per minute.

5. What did not work, stated plainly

The precision gain does not reach the trigger. R2 has clearly better lane precision than R0 (0.553 vs 0.494, 15/15 folds), yet at an equal false-alarm budget of 0.75 both deliver 67% detection. At a budget of 1.0 the 7-class model is actually ahead (71% vs 66%). The retraining improved the classifier but not the trigger built on top of it.

Honest reading: the false-alarm reduction from 2.19 to 0.75 was achieved by the gating change — using Place and progress instead of raw Align — which required no retraining at all. The retrain is justified by the frame-level result (macro-F1 0.392 → 0.480) and by Place recovery, not by the trigger metric.

Threshold selection was corrected. An earlier sweep chose the operating threshold by inspecting test performance, which inflates the result. Choosing it per fold on the validation subjects instead gives 68% detection at 0.75 false alarms for the 7-class arm — essentially unchanged, so the earlier figure was not materially biased, but the leak-free procedure is the one reported.

6. Conclusions

- **Dropping the Lift class is the single largest improvement measured in this project.** macro-F1 0.392 → 0.476, winning on all 15 folds ($p = 0.0001$). The expectation that it would cost accuracy was wrong.
- **Capping pos_weight fixes the over-prediction it was meant to fix.** Lane precision 0.462 → 0.553, 15/15 folds, at no cost to macro-F1 — but the gain does not propagate to the trigger.
- **Focal loss is harmful here** and should not be used: -0.037 macro-F1, losing on 14 of 15 folds.
- **The trigger is now usable.** 0.75 false alarms per event at 67% detection, against 2.19 at 75% initially. The credit belongs to the gating change, not the retrain.
- **Place recovery is the practically important class result.** Recall 0.38 → 0.51 and precision 0.47 → 0.62 on the class that physically precedes the free-drive moment.
- **Remaining limit is unchanged:** 15 subjects, with each additional real participant worth about $+0.011$ macro-F1 and the learning curve not yet saturated. Synthetic copies do not substitute.

Appendix: configuration

Setting	Value
Model	GRU, 128 hidden, 1 layer
Window	120 frames (4 s), causal, hop 10
Features	235 columns → 251 (azimuth as sin/cos)
Targets	peak vector, multi-label BCE
Augmentation	mirror, one copy per recording
Validation	15-fold leave-one-subject-out
Classes	6 (Pull, Align, Screw, Connect, Clamp, Place)
pos_weight cap	2.0 (R2, R3)